ANALOG PULSE MODULATION / 2.1

What changes in a pulse?

Look at height, width and position.

> **Pulse modulation.** A method of representing a message by varying the amplitude, width or position of a pulse train according to the message.

A pulse occupies a short time interval. Its amplitude can be positive or negative. **Classification:** analog pulse modulation divides into PAM and **pulse-time modulation (PTM)**. PTM includes PWM (also PDM) and PPM. These remain analog methods because the varied parameter can take a continuous range of values.

PAM: **pulse-amplitude modulation**. PWM/PDM: **pulse-width/duration modulation**. PPM: **pulse-position modulation**.

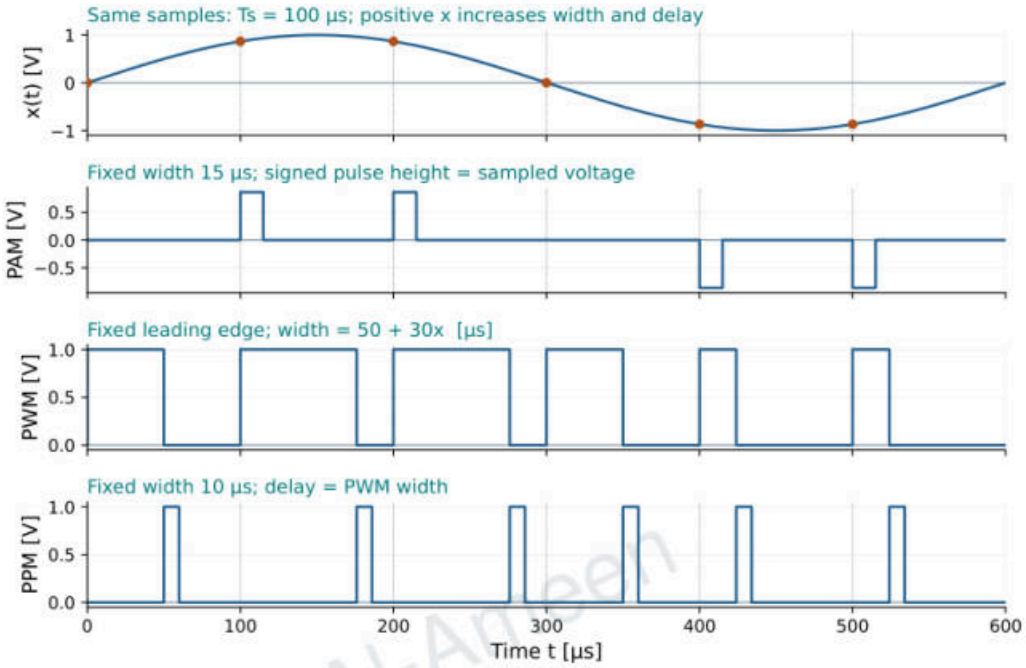


Figure 2.1: Aligned time axes: $T_s = 100\,\mu$s. PAM width is 15 μs. PWM width and PPM delay are $50 + 30x(kT_s)$ microseconds, with x in volts. PPM width is 10 μs.

Read the figure. The first curve is the message. In PAM, a larger sample gives a taller pulse. In PWM, it gives a wider pulse. In PPM, it gives a later pulse for the convention shown.

Types of analog pulse modulation

1. PAM: vary amplitude. Keep pulse width and nominal sampling times fixed.
2. PWM: vary width. Keep pulse height fixed.
3. PPM: vary position. Keep pulse height and width fixed.

ANALOG PULSE MODULATION / 2.2

Sampling: take a value at regular times

A sample is one measurement of the message.

Sampling. The process of measuring a continuous-time signal at selected instants to obtain a sequence of sample values.

Let $x(t)$ be the message. At time kT_s, measure its value. Write that value as $x(kT_s)$. Here k counts the samples: $0, 1, 2, \ldots$.

Two useful quantities. T_s is the time between samples. f_s is the number of samples taken each second.

$$\boxed{T_s = \frac{1}{f_s}}$$

Write f_m for the highest message frequency, as used in the sampling formulas. In the following exercises, $B = f_m$ is the same baseband limit. Let B be the highest frequency in the message, in hertz. To allow recovery of the message, use a sampling rate greater than twice B:

$$\boxed{f_s > 2B}$$

The theoretical boundary $2B$ is called the **Nyquist rate**. In practice, use a higher rate. Exactly $2B$ needs ideal conditions and care at the highest frequency.

Example 2.1 / A 3 kHz message

A message has $B = 3$ kHz. It is sampled at $f_s = 8$ kHz. Find the Nyquist rate and the time between samples.

Solution

Double the highest message frequency:

$$2B = 2(3) = 6\,\text{kHz}.$$

The selected rate is suitable because $8 > 6$ kHz.

Use the selected rate to find the actual time between samples:

$$T_s = \frac{1}{8000} = 0.000125\,\text{s} = 125\,\mu\text{s}.$$

Result. Take one sample every $125\,\mu$s.

Try it. If $f_s = 1000$ samples/s, how long is T_s?

Check: $T_s = 1/1000$ s $= 1$ ms.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.3

Find the Nyquist rate from the formula

Convert radians per second to hertz before doubling.

For $\cos(\omega t)$ or $\sin(\omega t)$, the frequency is $f = \omega/(2\pi)$ Hz. The Nyquist rate is $f_N = 2f_m$ and the Nyquist interval is $T_N = 1/f_N$.

Example 2.2 / A sum of three frequency components

$$x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t).$$

Divide each angular frequency by 2π:

$$f_1 = 25\,\text{Hz}, \qquad f_2 = 150\,\text{Hz}, \qquad f_3 = 50\,\text{Hz}.$$

The largest is $f_m = 150$ Hz. Therefore

$$f_N = 2(150) = 300\,\text{samples/s}, \qquad T_N = 1/300 = 3.333\,\text{ms}.$$

The coefficients 3, 10 and -1 affect amplitudes, not these frequencies.

Example 2.3 / A product of two cosines

$$x(t) = \frac{1}{2\pi}\cos(4000\pi t)\cos(1000\pi t).$$

Use $2\cos A\cos B = \cos(A+B) + \cos(A-B)$:

$$x(t) = \frac{1}{4\pi}\left[\cos(5000\pi t) + \cos(3000\pi t)\right].$$

The frequencies are 2500 Hz and 1500 Hz. Choose the larger:

$$f_N = 2(2500) = 5000\,\text{samples/s}, \qquad T_N = 1/5000 = 0.2\,\text{ms}.$$

Explain why: Why do the frequency limits add when signals are multiplied?

Multiplication in time becomes convolution in frequency. For these cosines it creates the sum and difference frequencies explicitly.

Try it. Find the Nyquist rate for $6\cos(50\pi t) + 20\sin(300\pi t) - 10\cos(100\pi t)$.

Check: 300 samples/s. The frequencies are again 25, 150 and 50 Hz.

ANALOG PULSE MODULATION / 2.4

Natural PAM: let part of the message pass

The switch is ON for a short time.

Natural PAM. A form of pulse-amplitude modulation in which each pulse follows the message during the sampling gate's ON interval.

An ON switch passes the message. An OFF switch blocks it. During each ON interval, the output follows the message, even if the message changes slightly.

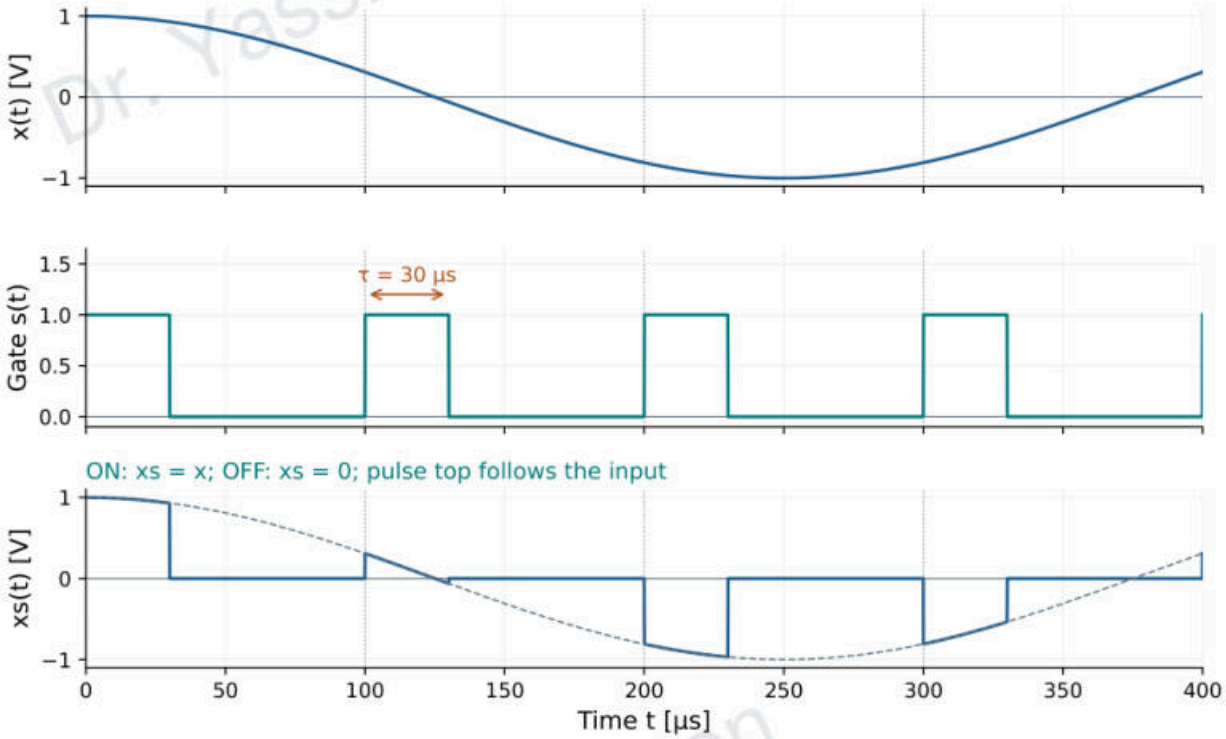


Figure 2.2: The gate decides which parts of the message appear at the output.

Read the figure. Choose one time. Read the gate in the middle panel. If it is one, the bottom signal equals the top signal. If it is zero, the bottom signal is zero. Let $s(t)$ be the gate and $x_s(t)$ the output. This operation is multiplication:

$$\boxed{x_s(t) = x(t)s(t)}$$

Let τ be the ON time of one pulse. The fraction of each period spent ON is the **duty factor** d:

$$\boxed{d = \frac{\tau}{T_s}}$$

For $\tau = 20\,\mu s$ and $T_s = 100\,\mu s$, $d = 20/100 = 0.2$. The switch passes the message for 20% of each period.

Try it. Is τ the same as d?

Check: No. τ is a time. d is a fraction and has no unit.

ANALOG PULSE MODULATION / 2.5

Flat-top PAM: keep one measured value

The top of each pulse stays horizontal.

Flat-top PAM. A form of pulse-amplitude modulation in which each pulse holds a constant amplitude equal to one message sample.

Measure the message once and keep that value for a short time. The output does not follow later changes until the next sample is taken.

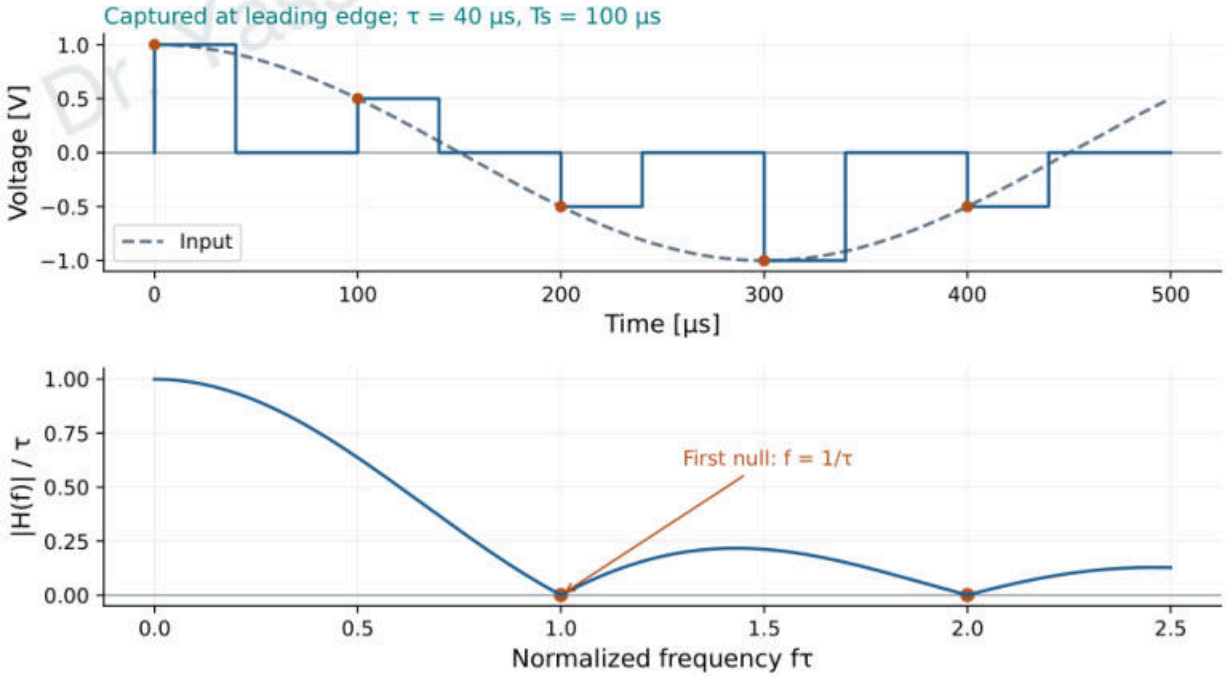


Figure 2.3: Top: held sample values, captured at the leading edge. Bottom: the magnitude response of a rectangular hold pulse, normalized by its width.

Read the upper figure first. Each orange dot is a measured value. The corresponding rectangle keeps that height. A negative sample gives a negative pulse.

The difference. Natural PAM follows the message during the ON interval. Flat-top PAM holds one value during that interval.

If the sample is 0.8 V, the pulse stays at 0.8 V for its full width. If the hold lasts until the next sample, the output becomes a staircase. The flat top gives a circuit time to measure the value. It does not remove all noise: noise can still change the measured height. The lower graph will be used later. It shows that holding a sample also changes its frequency content.

Try it. A message falls while a flat-top pulse is being held. Does the pulse top fall too?

Check: No. It stays at the captured value.

ANALOG PULSE MODULATION / 2.6

From an impulse to a flat-top pulse

A shift places the pulse. A sample value sets its height.

Let $x(t)$ be the message, $s(t)$ the ideal sampled signal and $g(t)$ the flat-top output. The symbol $*$ means convolution, not ordinary multiplication.

$$s(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s), \qquad h(t) = \begin{cases} 1, & 0 \le t < \tau, \\ 0, & \text{otherwise.} \end{cases}$$

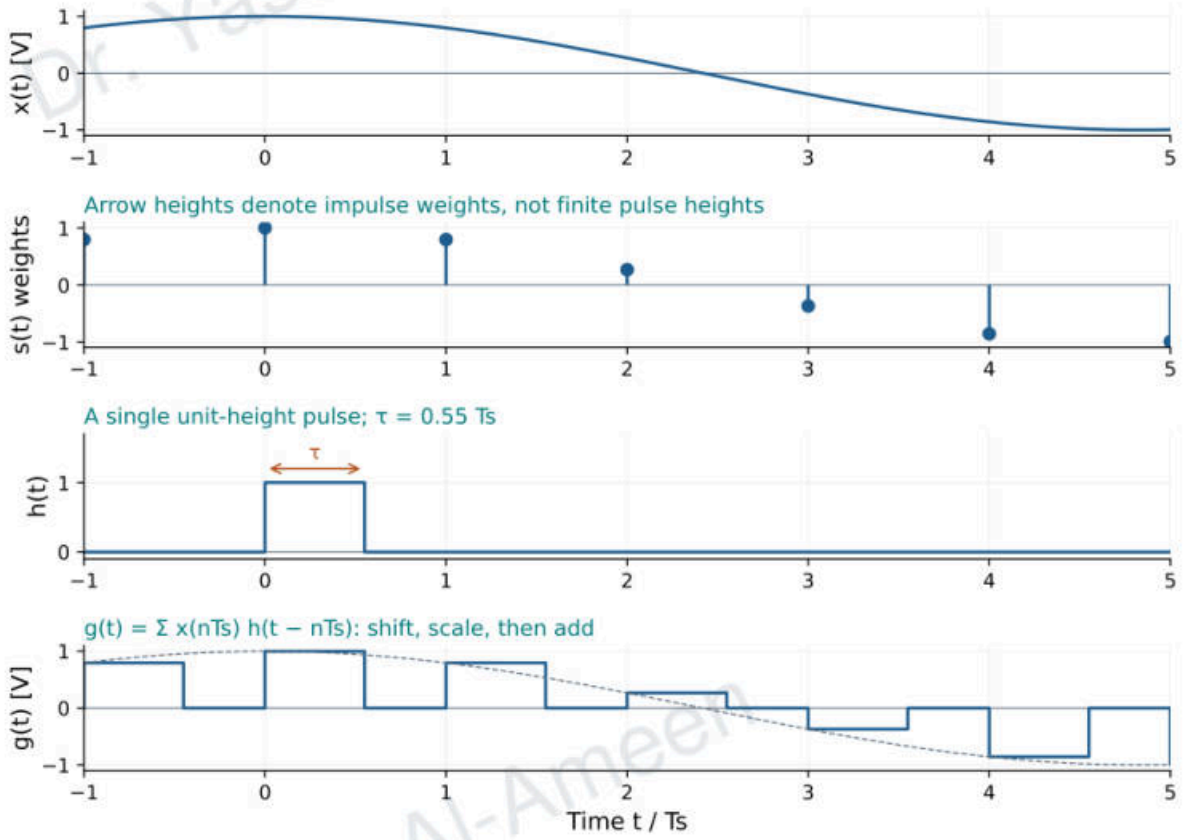


Figure 2.4: One rectangular kernel is shifted to each sampling instant and scaled by that sample.

Each impulse produces one rectangle: $x(nT_s)\delta(t - nT_s) * h(t) = x(nT_s)h(t - nT_s)$. Adding all rectangles gives

$$\boxed{g(t) = s(t) * h(t) = \sum_n x(nT_s)h(t - nT_s)}.$$

Explain why: Why is $h(t)$ drawn as one pulse?

It is the impulse response of the hold. The impulse train supplies the repetition. Convolving two periodic pulse trains is not the intended hold model.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.7

Sample, hold and reset

Understand the capacitor before studying the spectrum.

Sample-and-hold. A circuit that acquires an input voltage, stores it temporarily on a capacitor and makes the held voltage available to the next stage.

1. **Acquire:** close the sampling switch briefly. The capacitor approaches the input voltage.
2. **Hold:** open the sampling switch. A buffer reads the stored voltage with little loading.
3. **Reset, for return-to-zero PAM:** discharge after duration τ, then wait for the next sample. Do not short the input through both switches.

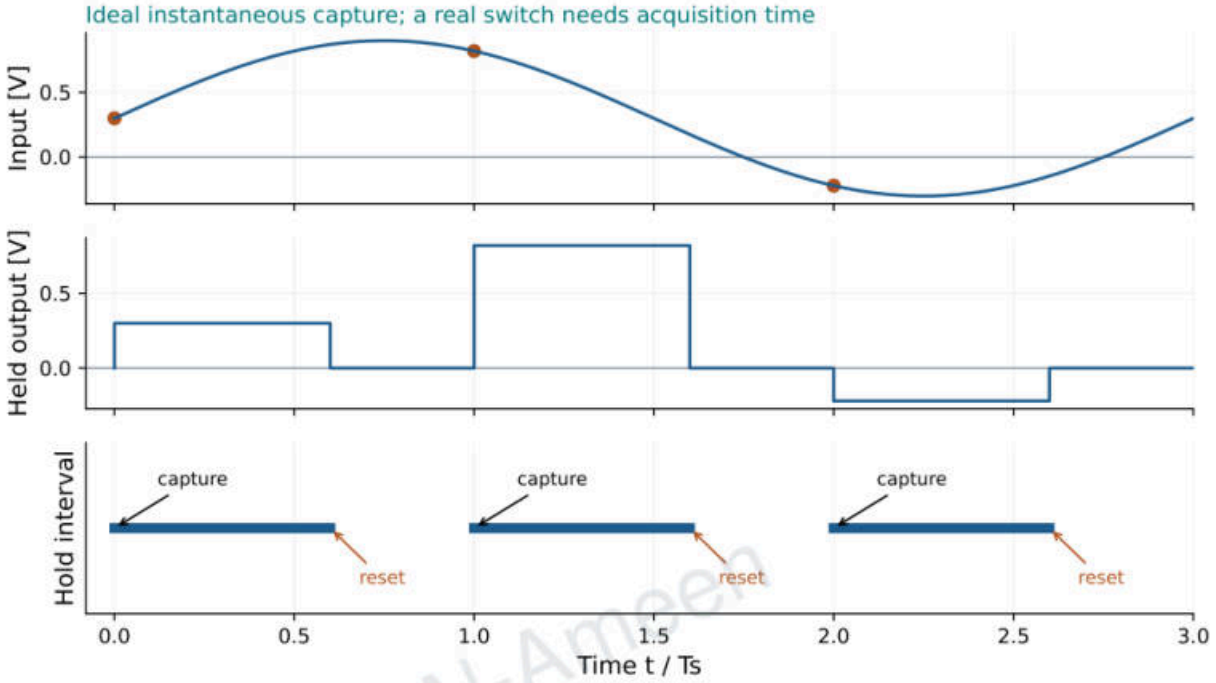


Figure 2.5: Ideal instantaneous capture, hold for $0.6T_s$, then reset. Real circuits need finite acquisition and reset intervals.

A zero-order hold keeps each sample until the next one, so it has no return-to-zero gap. A short flat-top PAM pulse returns to zero before the next sample.

Explain why: Does a flat top remove amplitude noise?

No. It gives a stable interval for reading the sample, but noise present at acquisition or added in transmission still changes the measured voltage.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.8

PAM: draw and explain the system

Follow the arrows and state the job of each block.

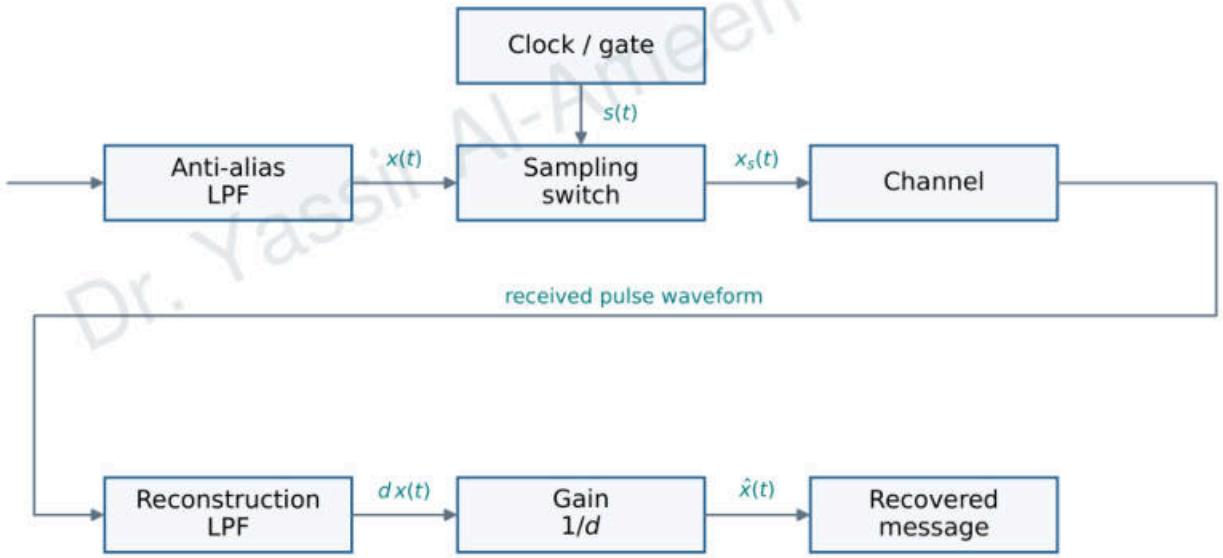


Figure 2.6: Natural-PAM transmitter and receiver. The receiver output after gain correction is the recovered message.

Functions of the blocks

1. **Anti-alias filter:** limits the message frequencies before sampling.
2. **Sampling switch:** passes the message while the gate is ON and blocks it while the gate is OFF.
3. **Receiver LPF:** keeps the central message-spectrum copy and rejects sampling copies.
4. **Gain correction:** multiplies by $1/d$ because the natural-PAM central copy is scaled by the duty factor d.

For flat-top PAM, use a sample-and-hold instead of the natural sampling switch. The receiver may also need aperture equalization to correct hold-related amplitude droop over the message band.

Questions and model answers

Explain why: Why is an anti-alias filter placed before the sampler?

It removes unwanted high-frequency components before they can overlap the message spectrum during sampling. Removing them afterward cannot undo aliasing in general.

Explain why: Why is the gain correction needed?

Natural sampling scales the central message copy by d. Multiplying by $1/d$ restores its original amplitude.

Draw and explain: Draw the natural-PAM transmitter and receiver. Label the clock/gate input and explain all four functions above.

ANALOG PULSE MODULATION / 2.9

A spectrum shows the frequencies

Sampling creates repeated copies of the spectrum.

The waveform $x(t)$ shows how a signal changes with time. Its spectrum $X(f)$ shows its frequency content. A message limited to B Hz has no content beyond $-B$ and B. For ideal impulse sampling, define $p_\delta(t) = \sum_n \delta(t - nT_s)$. Multiplication in time becomes convolution in frequency:

$$s(t) = x(t)p_\delta(t), \qquad S(f) = X(f) * \left[f_s \sum_n \delta(f - nf_s) \right] = f_s \sum_n X(f - nf_s).$$

The impulse $\delta(t)$ has unit area. A shifted impulse in frequency shifts the entire message spectrum. After sampling, copies appear at 0, $\pm f_s$, $\pm 2f_s$ and so on. A low-pass filter (LPF) keeps the central copy and rejects the others.

Example 2.4 / Find a gap for the filter

Use $B = 3$ kHz and $f_s = 8$ kHz.

The central copy ends at 3 kHz. The next copy is centred at 8 kHz and begins at

$$f_s - B = 8 - 3 = 5\,\text{kHz}.$$

There is a gap from 3 to 5 kHz. A filter cutoff of 4 kHz lies in that gap.

In general, choose a cutoff f_c between the message edge and the next copy:

$$\boxed{B < f_c < f_s - B}.$$

If the copies overlap, their values add together. A filter cannot simply separate them again.

Aliasing. Distortion caused when sampling makes different frequency components indistinguishable, typically because repeated spectra overlap.

How to avoid aliasing. Limit the input frequencies before sampling, and choose a sampling rate greater than twice the message bandwidth.

Try it. For $B = 2$ kHz and $f_s = 6$ kHz, where is the gap?

Check: Between 2 and 4 kHz.

Explain why: Why is the practical sampling rate chosen above twice the bandwidth?

It separates adjacent spectral copies and leaves a transition band for a realizable reconstruction filter.

ANALOG PULSE MODULATION / 2.10

Aliasing: watch the copies overlap

Compare the spectral support, not just the number of samples.

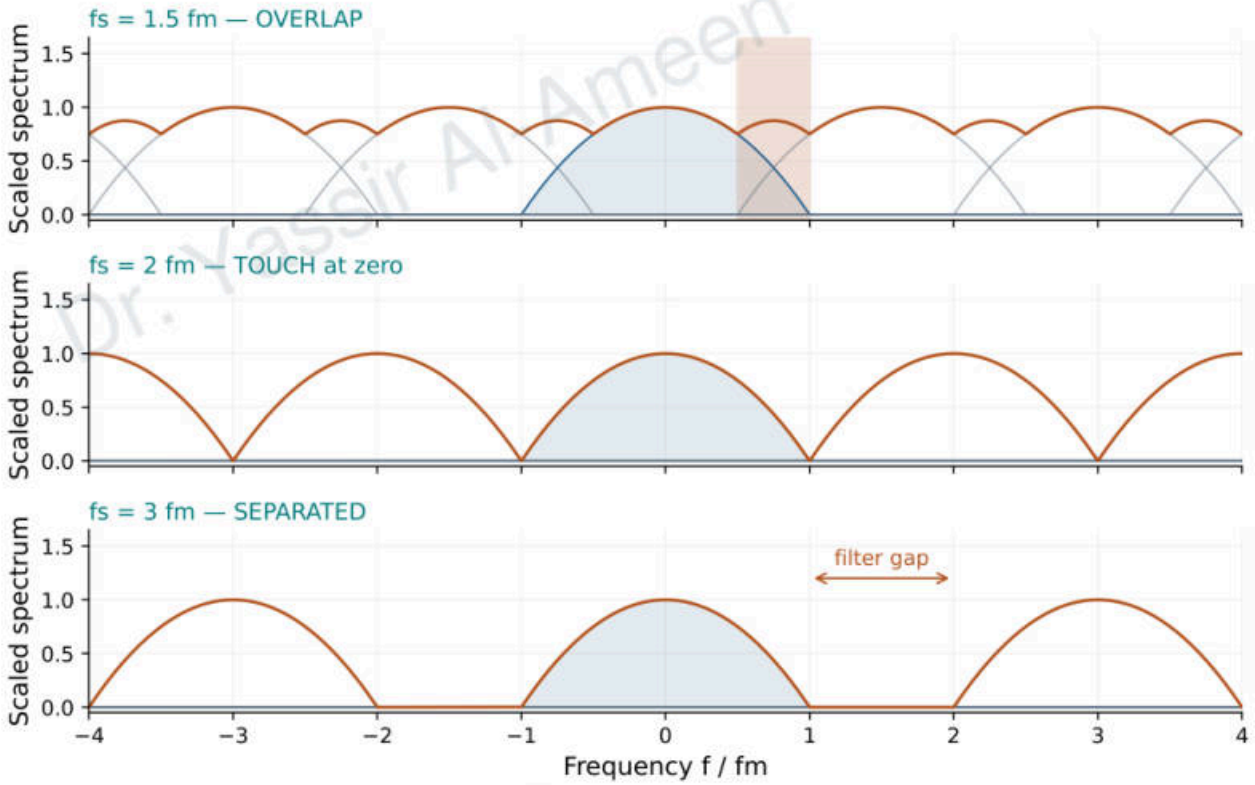


Figure 2.7: The same bandlimited spectrum sampled below, at and above $2f_m$. Orange is the sum of the repeated copies.

1. $f_s < 2f_m$: neighbouring copies overlap. Their values add in the shaded frequency intervals.
2. $f_s = 2f_m$: the supports touch. Recovery requires ideal filtering and suitable behaviour at the endpoints.
3. $f_s > 2f_m$: a gap of $f_s - 2f_m$ separates adjacent supports. This gap is in frequency, not a gap between time samples.

Example 2.5 / A 100 Hz cosine sampled at 150 Hz

The discrete samples are $\cos(2\pi 100n/150)$. Subtract one full cycle per sample:

$$\cos(2\pi 100n/150) = \cos(-2\pi 50n/150) = \cos(2\pi 50n/150).$$

Thus 100 Hz and 50 Hz give the same samples. Without prior band information, the receiver cannot choose between them.

ANALOG PULSE MODULATION / 2.11

Reconstruction: add weighted sinc pulses

Every sample contributes to the continuous waveform.

Interpolation. Reconstructing values between sample instants from the available samples under a specified signal model.

For a signal bandlimited below $f_s/2$, ideal reconstruction can be written

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\,\mathrm{sinc}\left(\frac{t - nT_s}{T_s}\right).$$

Here $\mathrm{sinc}(u) = \sin(\pi u)/(\pi u)$ and its value at zero is one.

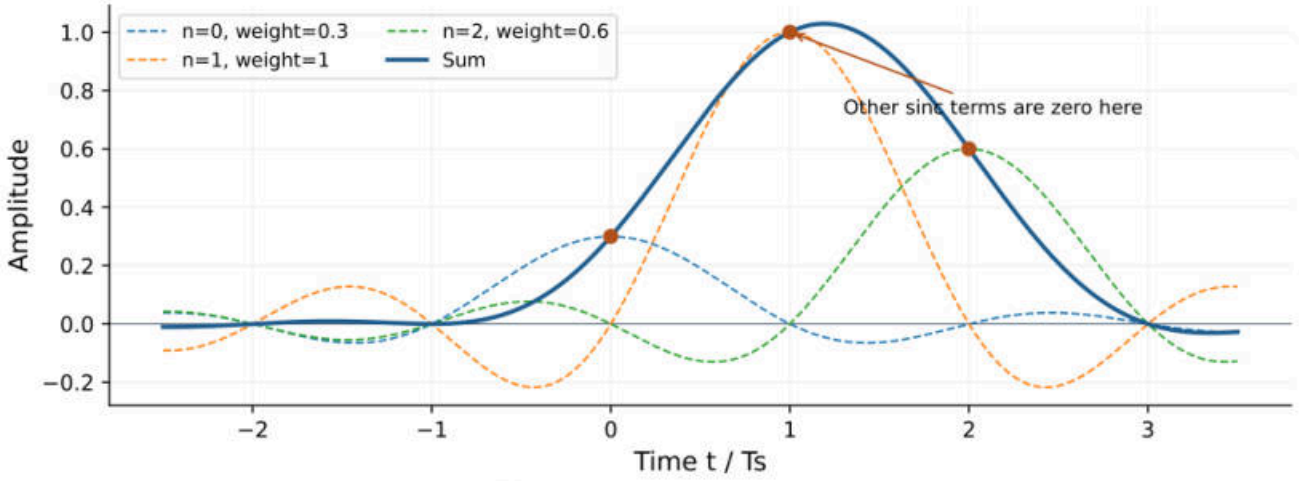


Figure 2.8: An illustrative bandlimited waveform made from three weighted sinc functions. The orange dots are the three nonzero sample values.

1. Place a sinc pulse at each sample time nT_s.
2. Multiply that sinc by the sample value $x(nT_s)$.
3. Add the contributions. At any sampling instant, all the other sinc terms are zero.

For ideal impulse sampling, the spectrum is multiplied by $1/T_s$. The ideal reconstruction filter therefore has gain T_s in its passband. A practical filter uses a finite transition band and approximates this response.

Explain why: Why is merely joining the dots not exact sinc reconstruction?

Straight-line interpolation uses triangular basis functions. Ideal bandlimited reconstruction uses sinc functions, which extend over all time.

ANALOG PULSE MODULATION / 2.12

Natural-PAM spectrum: read the formula

One part sets the position. Another part sets the height.

A rectangular gate creates copies of $X(f)$. The copy at nf_s has a coefficient c_n, which sets its height. The integer n labels the copies. A negative coefficient reverses the sign of that spectral copy, so the signed spectrum can extend below zero.

$$\boxed{X_s(f) = \sum_{n=-\infty}^{\infty} c_n X(f - nf_s)}$$

Position: $X(f - nf_s)$ is the message spectrum moved to nf_s.

Height: c_n multiplies that copy. For centred rectangular gates,

$$c_n = d\,\mathrm{sinc}(nd), \qquad d = \tau/T_s.$$

The normalized sinc function is

$$\mathrm{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \qquad \mathrm{sinc}(0) = 1.$$

Its value is zero at nonzero integers: $\pm 1, \pm 2, \ldots$.

Example 2.6 / Recover the central copy

Use $d = 0.2$. The central copy has $n = 0$, so

$$c_0 = 0.2\,\mathrm{sinc}(0) = 0.2.$$

The LPF keeps $0.2X(f)$. Multiply by $1/0.2 = 5$ to restore $X(f)$.

Where does the coefficient come from?

For a gate of height one, add its Fourier contributions over the ON interval and divide by the period:

$$c_n = \frac{1}{T_s}\int_{-\tau/2}^{\tau/2} e^{-j2\pi n f_s t}\,dt = \frac{\tau}{T_s}\,\mathrm{sinc}\left(\frac{n\tau}{T_s}\right).$$

Here $j^2 = -1$. The integral produces the same coefficient used above.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.13

Two natural-PAM examples

Find the copy positions before calculating their heights.

Let the message spectrum be one between $-B$ and B, and zero outside. Sample it at $f_s = 4B$.

Example 2.7 / Pulse width $T_s/3$

Find the duty factor and coefficient.

$$d = \frac{1}{3}, \qquad c_n = \frac{1}{3}\,\mathrm{sinc}(n/3)$$

Evaluate the first three heights.

$$c_0 = 0.3333, \quad c_1 = 0.2757, \quad c_2 = 0.1378$$

The first zero is at sinc argument one.

$$n/3 = 1 \Rightarrow n = 3, \qquad 3f_s = 12B$$

Example 2.8 / Pulse width $T_s/2$

Find the coefficient.

$$d = \frac{1}{2}, \qquad c_n = \frac{1}{2}\,\mathrm{sinc}(n/2)$$

Find the first missing copy.

$$n/2 = 1 \Rightarrow n = 2, \qquad 2f_s = 8B$$

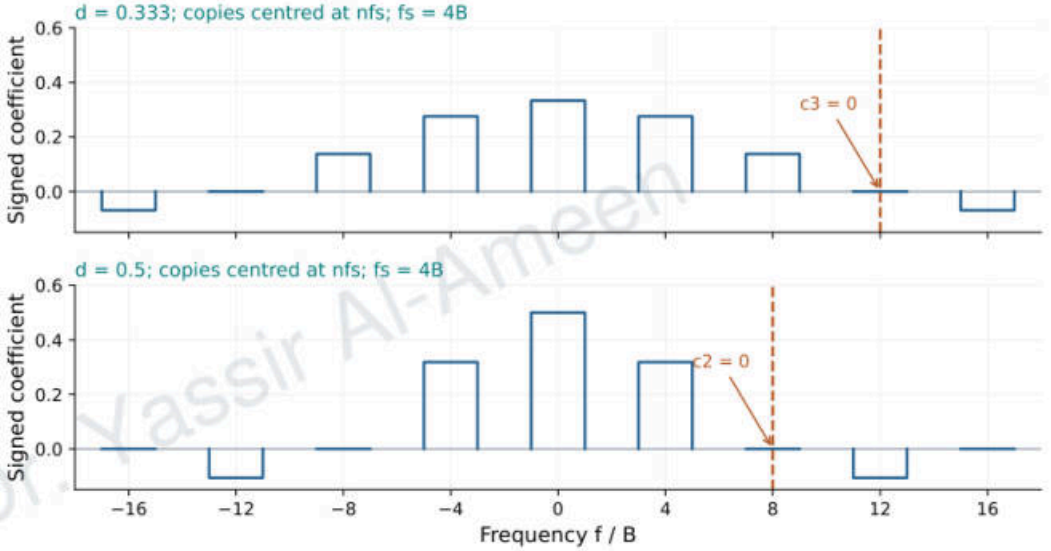


Figure 2.9: Compare the heights and the first missing copy in the two cases.

Read the figure. Each copy is centred a further $4B$ from the previous one. The orange line marks a missing copy. Copies beyond that line can still exist. A missing copy is not the end of the spectrum. Ideal rectangular pulses do not have a finite spectral limit.

ANALOG PULSE MODULATION / 2.14

Why does holding change the signal?

A long hold reduces high-frequency components more.

A hold pulse has a fixed duration τ. Its frequency response is

$$H(f) = \tau \operatorname{sinc}(\tau f)$$

for a rectangle centred at zero. The quantity $H(f)$ tells us how much the hold changes each frequency component. At low frequency, the normalized factor $H(f)/\tau$ is close to one. At a higher frequency, it becomes smaller.

Aperture effect. The frequency-dependent amplitude reduction caused by the finite duration of a flat-top sampling pulse.

Example 2.9 / Find the loss at 3 kHz

Use $f = 3000$ Hz and $\tau = 50\,\mu$s.

First multiply frequency by hold time:

$$f\tau = 3000(50 \times 10^{-6}) = 0.15.$$

Put this value into the sinc function:

$$\operatorname{sinc}(0.15) = \frac{\sin(0.15\pi)}{0.15\pi} = 0.9634.$$

The remaining amplitude is 96.34%. The reduction is $100 - 96.34 = 3.66\%$.

If the answer is required in decibels, use

$$20 \log_{10}(0.9634) = -0.324\,\text{dB}.$$

Connect this to the spectrum formula

Ideal sampling makes repeated copies. Holding multiplies all of them by $H(f)$:

$$X_s(f) = \frac{1}{T_s} H(f) \sum_k X(f - kf_s).$$

The LPF keeps the central copy. Correcting its constant gain still leaves the sinc-shaped reduction. An equalizer can compensate within a useful frequency band. For a causal hold, $h(t) = 1$ on $[0, \tau)$,

$$H(f) = \int_0^\tau e^{-j2\pi ft}\, dt = \tau \operatorname{sinc}(f\tau) e^{-j\pi f\tau}.$$

The exponential represents delay $\tau/2$. The sinc factor causes amplitude droop. With the notation $g(t) = s(t) * h(t)$, the output spectrum is $G(f) = S(f)H(f)$. Do not confuse this with the product $x(t)p(t)$ used for natural sampling.

ANALOG PULSE MODULATION / 2.15

Pulse-width calculations

Find the period, then the width, then the requested estimate.

The following exercises use the timing estimate

$$\boxed{B_{\text{est}} = \frac{1}{2\tau}}.$$

This estimate is different from the first zero of a rectangular-pulse spectrum, which is $1/\tau$. Neither is an exact upper frequency limit of the complete pulse spectrum.

Example 2.10 / $f_s = 4B$ and $\tau = T_s/3$

Find the period: $T_s = 1/(4B)$.

Find the pulse width: $\tau = T_s/3 = 1/(12B)$.

Use the requested estimate:

$$B_{\text{est}} = 1/(2\tau) = 6B.$$

The first pulse zero is $1/\tau = 12B$.

Example 2.11 / Voice signal

Use $f_s = 8$ kHz and $\tau = 0.1T_s$ for a 3 kHz message.

Find the period.

$$T_s = \frac{1}{8000} = 125\,\mu\text{s}$$

Take one tenth of the period.

$$\tau = 0.1(125) = 12.5\,\mu\text{s}$$

Convert to seconds and substitute.

$$B_{\text{est}} = \frac{1}{2(12.5 \times 10^{-6})} = 40\,\text{kHz}$$

Try it. Is 40 kHz the original voice bandwidth?

Check: No. The voice bandwidth is 3 kHz. The 40 kHz value is the specified pulse-width estimate.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.16

Two more width examples

The duty factor is a fraction of the period.

Example 2.12 / Sinusoidal message

Let $g(t) = \sin(2\pi t/T_m)$, with $T_m = 0.2$ s. Use $f_s = 100$ Hz and $\tau = 0.25T_s$.

The message frequency is $1/0.2 = 5$ Hz. The selected sampling rate is greater than twice this value.

Find the sampling period:

$$T_s = 1/100 = 0.01\,\text{s}.$$

The pulse lasts one quarter of the period:

$$\tau = 0.25(0.01) = 0.0025\,\text{s}.$$

Find the timing estimate:

$$B_{\text{est}} = 1/[2(0.0025)] = 200\,\text{Hz}.$$

Example 2.13 / Ten-percent duty factor

Use $f_s = 150$ Hz and $d = 0.1$.

Convert the duty factor to a duration:

$$\tau = dT_s = d/f_s = 0.1/150 = 1/1500\,\text{s}.$$

Then calculate the requested estimate:

$$B_{\text{est}} = 1/(2\tau) = 1500/2 = 750\,\text{Hz}.$$

Try it. Use $f_s = 200$ Hz and $d = 0.2$. Find the pulse width and timing estimate.

Check: $\tau = 0.2/200 = 1$ ms. Then $B_{\text{est}} = 500$ Hz.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.17

PWM: change the pulse width

A larger message value gives a wider pulse.

PWM. Pulse-width modulation varies pulse duration according to the message while keeping pulse amplitude fixed.

Begin with a width τ_0 when the message is zero. Add a width proportional to the sample value:

$$\boxed{\tau_k = \tau_0 + k_w x(kT_s)}.$$

Here τ_k is the new width and k_w is the width change per volt. The model requires $0 < \tau_k < T_s$. The equivalent form is $\tau_k = \tau_0[1 + \mu x(kT_s)]$, where $k_w = \tau_0\mu$. If x is in volts, μ has units V^{-1}.

Example 2.14 / Convert voltage to width

Use $\tau_0 = 50\,\mu\mathrm{s}$ and $k_w = 20\,\mu\mathrm{s/V}$.

Write the width law.

$$\tau_k = \tau_0 + k_w x(kT_s)$$

For a positive sample of 1.5 V:

$$\tau_k = 50 + 20(1.5) = 80\,\mu\mathrm{s}$$

For a zero sample:

$$\tau_k = 50 + 20(0) = 50\,\mu\mathrm{s}$$

For a negative sample of -1.5 V:

$$\tau_k = 50 + 20(-1.5) = 20\,\mu\mathrm{s}$$

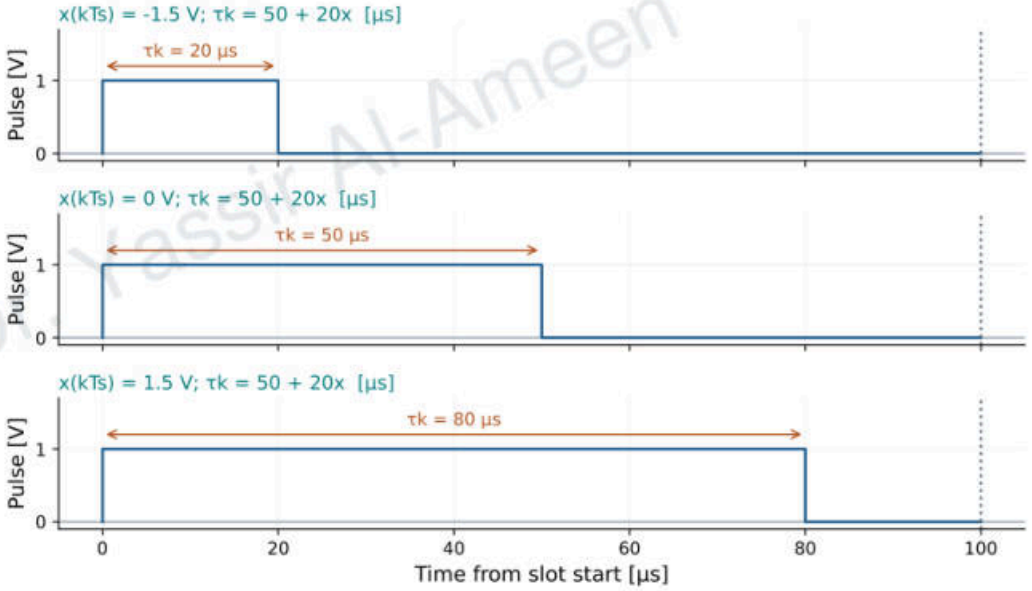


Figure 2.10: The same pulse height with widths of 20, 50 and 80 microseconds.

Check the result. Each pulse must end before the next slot begins. With $T_s = 100\,\mu\mathrm{s}$, all three widths fit.

ANALOG PULSE MODULATION / 2.18

Three PWM timing conventions

The width law stays the same. The fixed reference changes.

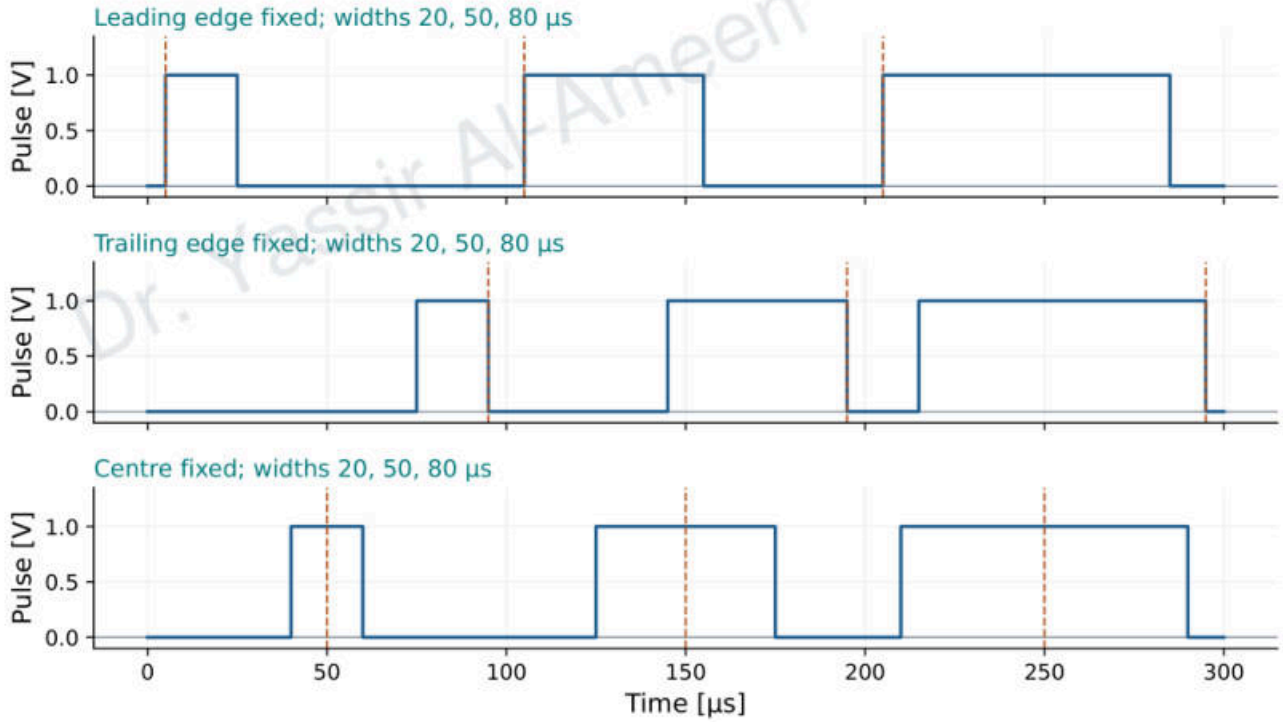


Figure 2.11: Widths of $0.2T_s$, $0.5T_s$ and $0.8T_s$ under three edge conventions. Orange marks the fixed reference in each slot.

1. **Fixed leading edge:** the start is fixed and the trailing edge moves.
2. **Fixed trailing edge:** the end is fixed and the leading edge moves.
3. **Fixed centre:** both edges move equally away from the centre as the pulse widens.

The examples use a fixed leading edge. A positive message increases width. Negative messages decrease width relative to the zero-message width τ_0.

Explain why: Can the same message produce different-looking PWM drawings?

Yes. The drawing depends on which edge or centre is fixed and on the sign of the voltage-to-width mapping. State the convention before interpreting it.

Try it. A centred pulse has width $40\,\mu$s and centre at $60\,\mu$s. Find its edges.

Check: Start $= 40\,\mu$s and end $= 80\,\mu$s.

ANALOG PULSE MODULATION / 2.19

How the PWM circuit works

A ramp catches up with the control voltage.

A ramp rises steadily from zero. A comparator compares the ramp with a held control voltage. The output stays high while the control voltage is higher than the ramp. When the ramp reaches it, the output falls.

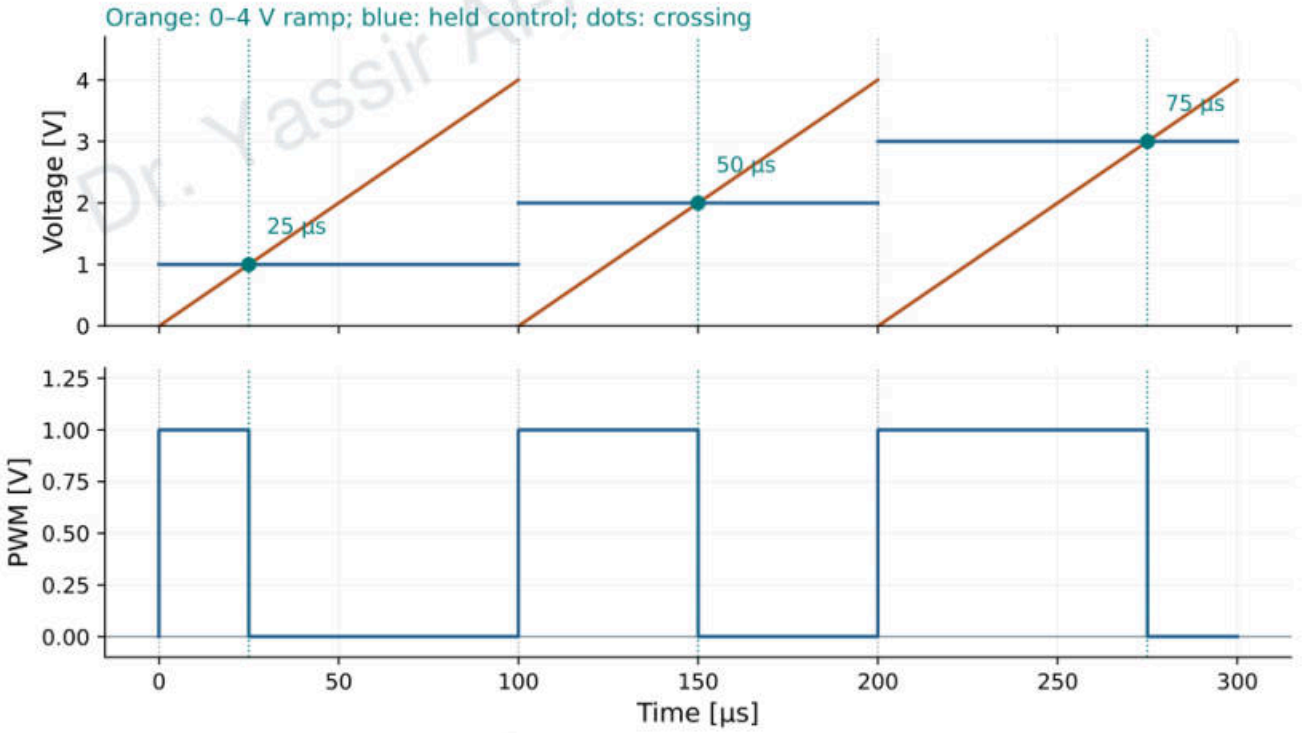


Figure 2.12: The crossing in the top panel determines the pulse end in the bottom panel.

Read the first slot. The control is one quarter of the ramp peak. The ramp reaches it after one quarter of the period. The PWM pulse therefore lasts $0.25T_s$. Let the ramp peak be V_r and the held control voltage be v_k. The width is

$$\boxed{\tau_k = T_s \frac{v_k}{V_r}}.$$

For $T_s = 100\,\mu$s, $V_r = 4$ V and $v_k = 3$ V, the pulse width is $100(3/4) = 75\,\mu$s. To allow negative message samples, first add an offset to make the control voltage positive. For the preceding width law, use $v_k = (V_r/T_s)[\tau_0 + k_w x(kT_s)]$, with $0 < v_k < V_r$.

ANALOG PULSE MODULATION / 2.20

PWM: generation and detection blocks

Convert voltage to time, then convert time back to voltage.

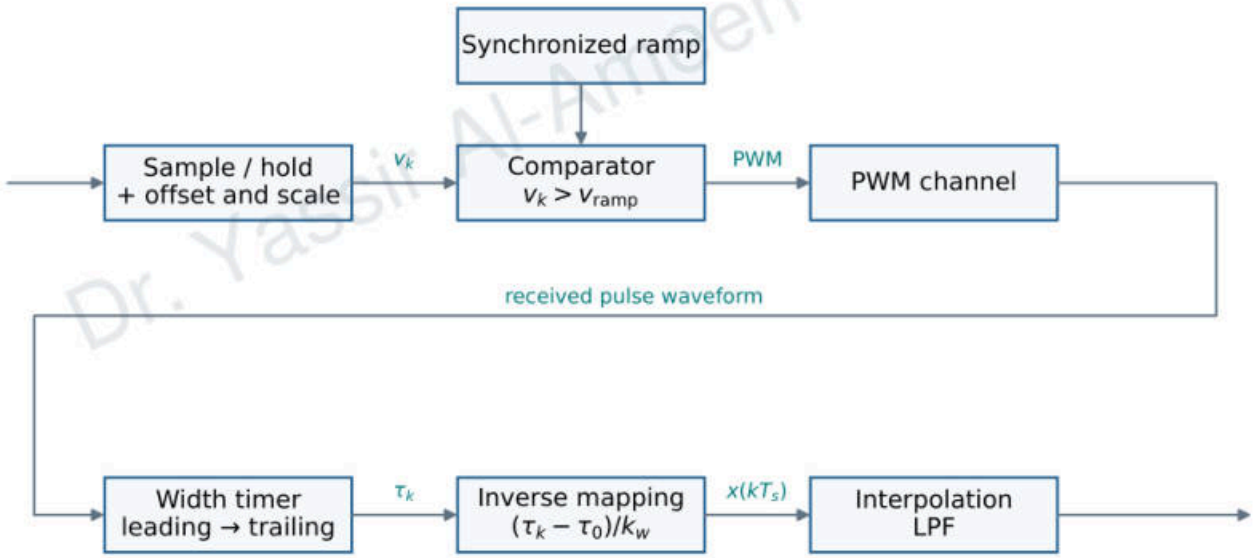


Figure 2.13: A sampled PWM implementation with an explicit width-measuring receiver. The input ramp and message samples use the same slot timing.

Functions of the blocks

1. **Sample-and-hold with offset/scale:** keeps one control voltage fixed during each slot and places it inside the ramp range.
2. **Comparator:** stays high while the control voltage exceeds the ramp. The crossing determines the pulse width.
3. **Width timer:** measures the time from the rising edge to the falling edge of the received pulse.
4. **Inverse mapping:** subtracts τ_0 and divides by k_w to recover the message sample.
5. **Interpolation filter:** reconstructs a continuous message from the recovered samples under the sampling conditions.

$$x(kT_s) = \frac{\tau_k - \tau_0}{k_w}.$$

Explain why: Why does a higher control voltage make the pulse wider?

The rising ramp needs more time to reach a higher control voltage. The comparator therefore remains high for longer.

Explain why: Why must the pulse width remain below the slot duration?

The pulse must finish before the next slot begins. Otherwise adjacent pulses can merge and the receiver cannot measure each width independently.

ANALOG PULSE MODULATION / 2.21

Recover PWM with a ramp and hold

Convert pulse duration into a measurable voltage.

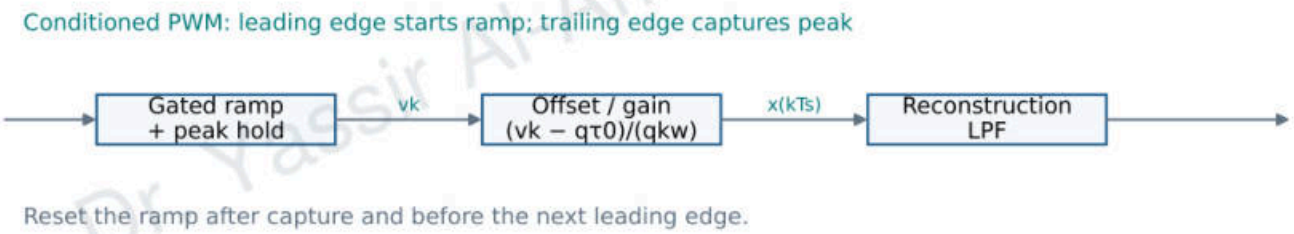


Figure 2.14: A conceptual analog width-to-amplitude detector. Its clock and reset controls must be coordinated.

1. Condition the received edges with a comparator or Schmitt trigger. This reduces sensitivity to amplitude fluctuations, but does not remove all timing error.
2. Start a ramp of slope q at the leading edge. Stop and capture it at the trailing edge. Its peak is $v_k = q\tau_k$.
3. Subtract the zero-message peak $q\tau_0$ and divide by qk_w to recover $x(kT_s)$.
4. Reset the ramp for the next pulse and reconstruct the message from the held samples.

$$\boxed{x(kT_s) = \frac{v_k - q\tau_0}{qk_w}}.$$

Example 2.15 / Convert width to voltage and back

Use $q = 0.05\ \text{V}/\mu\text{s}$, $\tau_0 = 50\ \mu\text{s}$, $k_w = 20\ \mu\text{s/V}$ and $\tau_k = 80\ \mu\text{s}$.

Find the captured voltage.

$$v_k = q\tau_k = 0.05(80) = 4\,\text{V}$$

Find the zero-message voltage.

$$q\tau_0 = 0.05(50) = 2.5\,\text{V}$$

Remove the offset and divide by the sensitivity.

$$x(kT_s) = \frac{v_k - q\tau_0}{qk_w} = \frac{4 - 2.5}{0.05(20)} = 1.5\,\text{V}$$

Explain why: Does PWM width measurement need an external slot-start reference?

Not necessarily. Both edges are present in the pulse itself. The receiver still needs correct edge association and suitable timing for capture and reset. PPM, in contrast, needs a separate or recovered position reference.

ANALOG PULSE MODULATION / 2.22

PPM: move a short pulse

The pulse width stays fixed. Its starting time changes.

PPM. Pulse-position modulation varies the delay of a fixed-amplitude, fixed-width pulse relative to a timing reference according to the message.

Take the end of each PWM pulse and start a short pulse there. Keep the new pulse width fixed. The result is PPM.

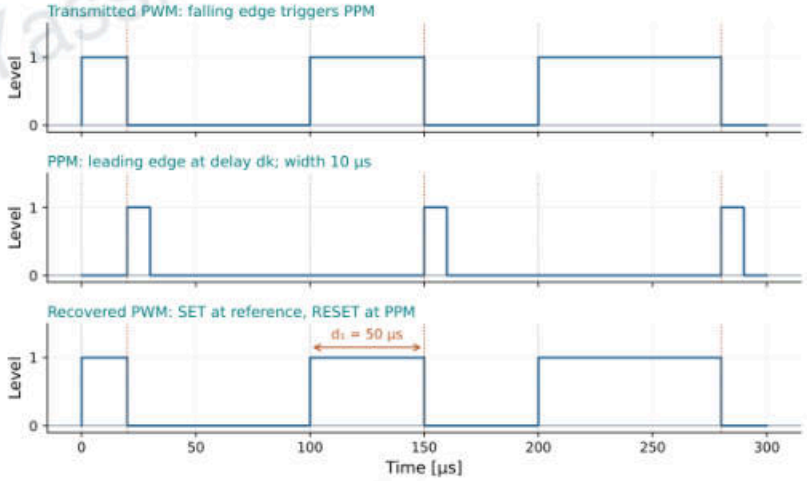


Figure 2.15: PWM ends, PPM begins, and the receiver measures the delay.

Read one dotted line. It passes through the end of a PWM pulse and the start of the PPM pulse. A wider PWM pulse creates a later PPM pulse. At the receiver, start a timer at the beginning of the slot. Stop it when the PPM pulse arrives. Call this measured delay d_k.

$$\boxed{x(kT_s) = \frac{d_k - \tau_0}{k_w}}.$$

Example 2.16 / Recover the sample

Use $d_k = 80\,\mu\text{s}$, $\tau_0 = 50\,\mu\text{s}$ and $k_w = 20\,\mu\text{s/V}$.

Write the recovery law.

$$x(kT_s) = \frac{d_k - \tau_0}{k_w}$$

Remove the initial delay.

$$d_k - \tau_0 = 80 - 50 = 30\,\mu\text{s}$$

Divide by the delay per volt.

$$x(kT_s) = \frac{30}{20} = 1.5\,\text{V}$$

For fixed PPM pulse width τ_p, require $0 \le d_k$ and $d_k + \tau_p < T_s$ so the whole pulse fits its slot. The receiver needs an accurate slot reference. If that reference is wrong, the measured delay is wrong too.

ANALOG PULSE MODULATION / 2.23

PPM: generation and detection blocks

The receiver must know when each slot starts.

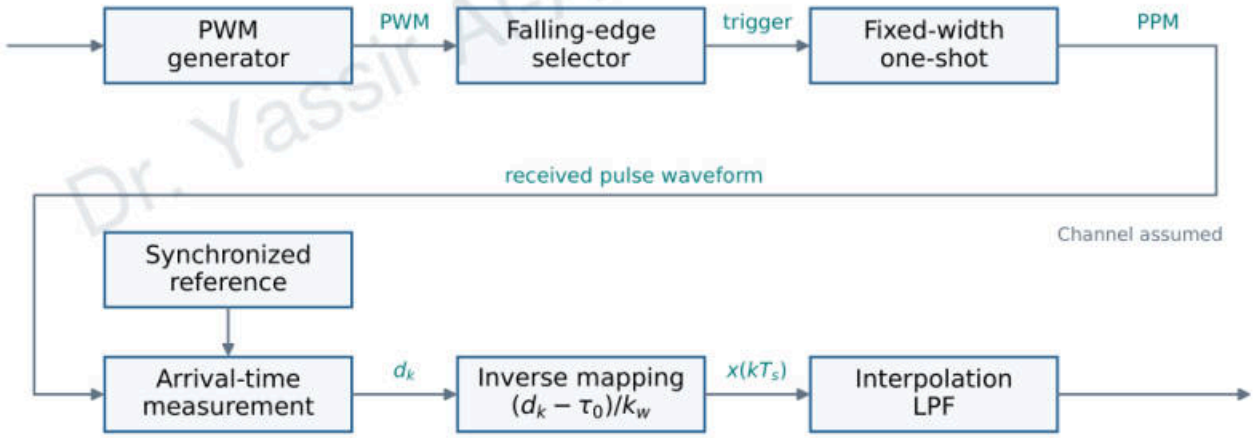


Figure 2.16: PPM generation from a PWM falling edge and recovery by measuring arrival time relative to a synchronized reference.

Functions of the blocks

1. **PWM generator:** makes a pulse whose width represents the sample.
2. **Falling-edge selector:** produces a trigger at the end of that pulse.
3. **Fixed-width pulse generator:** produces one short pulse at each trigger. This is the PPM signal.
4. **Arrival-time measurement:** measures the delay from the known slot start to the received PPM leading edge.
5. **Inverse mapping and interpolation:** convert the delay to a sample value, then reconstruct the message.

Explain why: Why is synchronization essential in PPM?

The information is a delay relative to the slot start. Without an accurate reference, the receiver cannot distinguish a message-dependent delay from a clock error.

Explain why: Why is pulse width kept fixed in PPM?

Position carries the information. Keeping width and height fixed allows the receiver to read the message from timing alone.

Numerical check: For a delay of 70 μs, initial delay 50 μs and sensitivity 20 μs/V, the sample is $(70 - 50)/20 = 1$ V.

ANALOG PULSE MODULATION / 2.24

PPM receiver: convert position to width

A flip-flop measures the delay as a pulse duration.

Set-reset operation. A reference pulse sets the output high. The leading edge of the received PPM pulse resets it low. The high interval equals the measured delay.

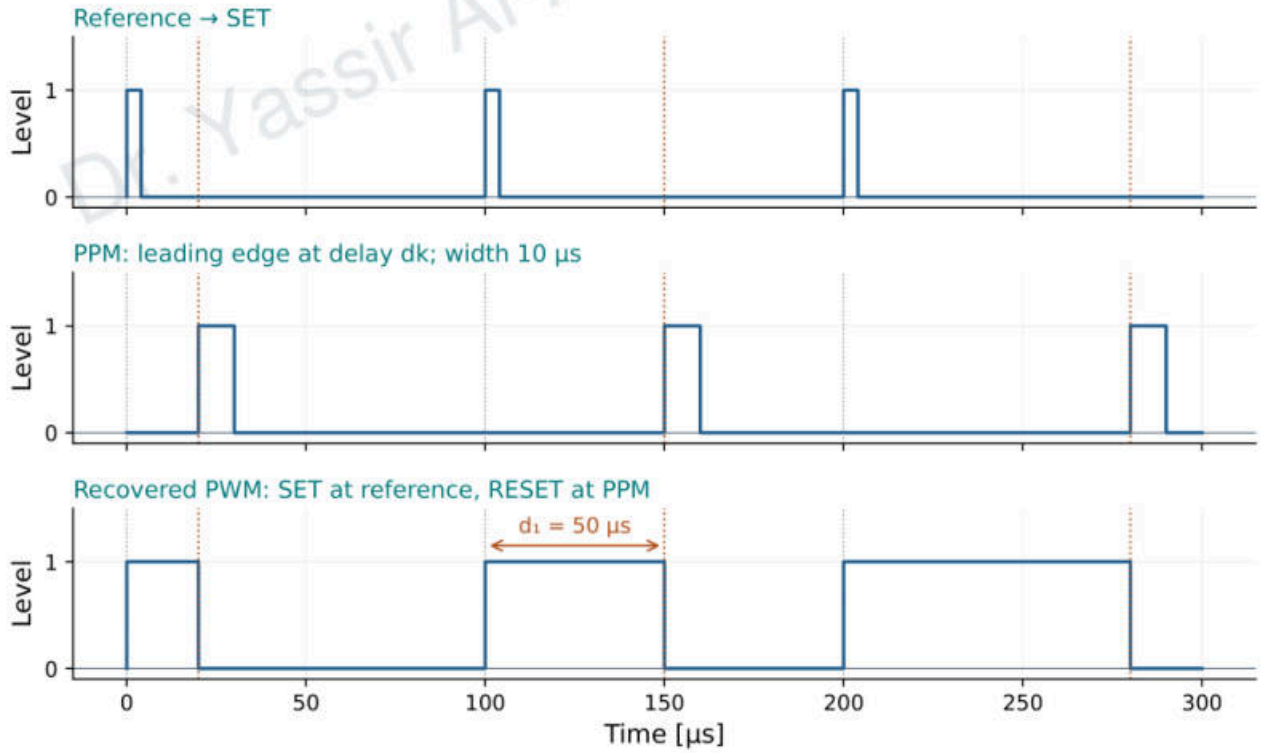


Figure 2.17: Reference pulses set the output. Delayed PPM pulses reset it. The resulting waveform is PWM.

1. Recover or supply a synchronized reference for the beginning of each slot.
2. Set the flip-flop at that reference instant.
3. Reset it at the received PPM leading edge. The resulting width is d_k.
4. Apply PWM demodulation and remove the offset to recover the message.

Explain why: Why must the reference arrive first in this implementation?

The circuit measures a nonnegative delay from SET to RESET. The transmitter offset and allowed message range must keep the PPM edge inside the intended slot.

Try it. The slot reference is 5 μs late. What happens to measured delay?

Check: It is 5 μs too short. With $k_w = 20\,\mu\text{s/V}$, the recovered value is biased by -0.25 V.

ANALOG PULSE MODULATION / 2.25

Noise: what does it change?

PAM reads height. PWM and PPM read time.

In PAM, extra voltage changes the measured pulse height. In PWM and PPM, noise can move the time at which the receiver detects an edge.

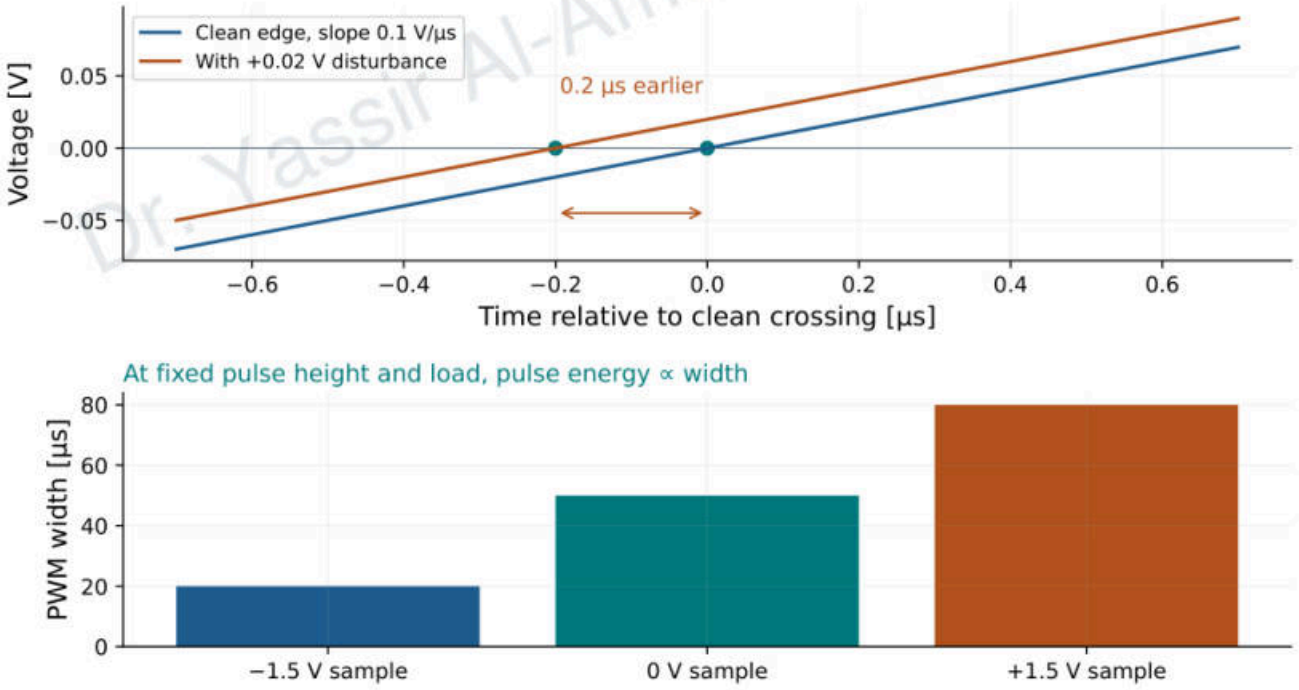


Figure 2.18: Noise can change an amplitude measurement or shift a timing measurement.

Read the upper graph. The clean edge crosses zero at the blue crossing time. Adding voltage moves the crossing to another time. The receiver then measures the wrong edge time. For a small voltage disturbance v_n and edge slope q, the approximate time error is

$$\Delta t \approx -\frac{v_n}{q}.$$

The slope is how quickly the voltage changes. A steeper edge usually gives a smaller time error for the same disturbance.

Example 2.17 / Estimate the time error

Use $v_n = 0.02$ V and $q = 0.1$ V/μs. The error magnitude is

$$|\Delta t| = 0.02/0.1 = 0.2\,\mu\text{s}.$$

An amplitude limiter can reduce height variation in PWM or PPM. It cannot restore an edge that has already moved in time.

Energy comparison. With the same load, a taller or wider pulse carries more energy. PAM can change energy by changing height. PWM changes it by changing width. PPM keeps height and width fixed, so moving the pulse does not change its energy.

ANALOG PULSE MODULATION / 2.26

Compare PAM, PWM and PPM

Use the changed pulse property to explain the differences.

Property	PAM	PWM	PPM
Carries the message	Pulse height	Pulse width	Pulse delay
Kept fixed	Width and nominal times	Height	Height and width
Receiver measures	Amplitude	Time between edges	Time from reference
Main noise concern	Height error	Edge-time error	Edge-time and reference errors
Energy per pulse	Can vary with height	Varies with width	Fixed for fixed height, width and load

Advantages and limitations

1. **PAM:** amplitude representation is direct and natural-PAM recovery can use an LPF and gain correction. Amplitude noise changes the sample value. Flat-top PAM also introduces aperture droop.
2. **PWM:** the fixed pulse height allows amplitude limiting. Accurate recovery still requires reliable edge timing, and pulse energy changes with width.
3. **PPM:** pulse height, width and energy can remain fixed. Reliable recovery requires accurate synchronization and edge detection.

Explain why: model answers

Explain why: Why does increasing the flat-top pulse width increase aperture distortion?

A longer hold makes the sinc response decrease more across the same useful baseband, provided that band lies in its first lobe. High-frequency message components are reduced more than low-frequency components.

Explain why: Why does moving a PPM pulse not change its energy?

Its amplitude and duration stay the same. Across the same load, energy depends on these quantities, not on the pulse's starting time.

Explain why: Why does amplitude limiting not remove all noise from PWM or PPM?

A limiter can reduce height variation, but it cannot generally restore the correct timing of an edge displaced by noise.

Compare: Explain three differences between PAM and PPM using the table. **Define:** State the meanings of PAM, PWM, PPM and aperture effect.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.27

Sinc problems: find the highest frequency

Use three rules instead of guessing from the waveform.

Use the normalized definition $\text{sinc}(at) = \sin(\pi at)/(\pi at)$. Thus the expression $\sin c(100\pi t)$, with $\sin c(u) = \sin(u)/u$, means $\text{sinc}(100t)$.

> **Three rules.** For $\text{sinc}(at)$ with $a \neq 0$, the bandwidth is $|a|/2$. When signals are added, use the larger bandwidth, unless cancellation occurs. When they are multiplied, their spectral support limits add.

The Nyquist rate is twice the highest frequency. The Nyquist interval is the inverse of that rate.

Example 2.18 / Five sampling-rate questions

Signal: $\text{sinc}(100t)$.

Halve the time-scaling factor.

$$B = 50\,\text{Hz}, \qquad f_N = 100\,\text{samples/s}, \qquad T_N = 10\,\text{ms}.$$

Signal: $\text{sinc}^2(100t)$.

For the square, add the two bandwidths.

$$B = 50 + 50 = 100\,\text{Hz}, \qquad f_N = 200\,\text{samples/s}, \qquad T_N = 5\,\text{ms}.$$

Signal: $\text{sinc}(100t) + \text{sinc}(50t)$.

For this sum, take the larger bandwidth.

$$B = \max(50, 25) = 50\,\text{Hz}, \qquad f_N = 100\,\text{samples/s}, \qquad T_N = 10\,\text{ms}.$$

Signal: $\text{sinc}(100t) + 3\,\text{sinc}^2(60t)$.

Find each bandwidth, then take the larger.

$$B = \max(50, 60) = 60\,\text{Hz}, \qquad f_N = 120\,\text{samples/s}, \qquad T_N = 8.333\,\text{ms}.$$

Signal: $\text{sinc}(100t)\,\text{sinc}(100t)$.

This is the same square as the second signal.

$$B = 50 + 50 = 100\,\text{Hz}, \qquad f_N = 200\,\text{samples/s}, \qquad T_N = 5\,\text{ms}.$$

Try it. Find the rate for $\text{sinc}(80t)\,\text{sinc}(40t)$.

Check: Bandwidth $= 40 + 20 = 60$ Hz. Rate $= 120$ samples/s.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.28

Recover a message from a pulse train

Find the duty factor first.

Example 2.19 / Width $1/(8B)$ and rate $2B$

The message is $g(t)$ and its bandwidth is B. The sampling pulses have height one and are centred on the sampling instants.

Invert the sampling rate:

$$T_s = \frac{1}{2B}.$$

Divide pulse width by period:

$$d = \frac{1/(8B)}{1/(2B)} = \frac{1}{4}.$$

The gate is ON for one quarter of each period. Therefore the central message copy is $g(t)/4$. Keep it with an ideal LPF and multiply it by four.

Write the full expression

Use $c_n = d\,\mathrm{sinc}(nd)$. For $d = 1/4$,

$$c_0 = \frac{1}{4}, \qquad c_n = \frac{\sin(n\pi/4)}{n\pi} \quad (n \neq 0).$$

Positive and negative harmonics combine into cosine terms. With $\omega_s = 2\pi f_s = 4\pi B$, the gate is

$$p_{T_s}(t) = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\left(\frac{n\pi}{4}\right) \cos(n\omega_s t).$$

Multiply each term by $g(t)$ to obtain the sampled signal:

$$\widetilde{g}(t) = \frac{1}{4} g(t) + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\left(\frac{n\pi}{4}\right) g(t) \cos(n\omega_s t).$$

The first term is the copy to keep. The other terms are shifted copies to reject. Recovery at exactly $2B$ assumes ideal filtering and no problematic spectral line at the band edge. A practical system uses a higher rate.

ANALOG PULSE MODULATION / 2.29

Two filters, then multiplication

Follow the signal path from left to right.



Figure 2.19: Filter each signal, then multiply the outputs.

Example 2.20 / Find the three Nyquist rates

The inputs are $g_1(t) = 10^4\,\mathrm{rect}(10^4 t)$ and $g_2(t) = \delta(t)$. Here $\delta(t)$ is a unit impulse. The filters are

$$H_1(\omega) = \mathrm{rect}\Big(\frac{\omega}{40000\pi}\Big), \quad H_2(\omega) = \mathrm{rect}\Big(\frac{\omega}{20000\pi}\Big).$$

The function $\mathrm{rect}(u)$ is one for $|u| < 1/2$ and zero outside. Thus the first cutoff is 20000π rad/s. Divide by 2π to convert to hertz:

$$B_1 = 20000\pi/(2\pi) = 10000\,\mathrm{Hz}.$$

The second cutoff is 10000π rad/s, giving

$$B_2 = 10000\pi/(2\pi) = 5000\,\mathrm{Hz}.$$

These inputs fill the filter passbands. Double each output bandwidth:

$$f_{N,1} = 20000\,\mathrm{samples/s}, \qquad f_{N,2} = 10000\,\mathrm{samples/s}.$$

The outputs are multiplied. Their spectral support limits add:

$$B_y = 10000 + 5000 = 15000\,\mathrm{Hz}.$$

Double this value to find the final rate:

$$\boxed{f_{N,y} = 30000\,\mathrm{samples/s}}.$$

Try it. Would adding the outputs give the same bandwidth?

Check: No. Here addition gives 10 kHz. Multiplication gives 15 kHz.

ANALOG PULSE MODULATION / 2.30

Compare three sampling rates

Use the same signal each time.

Take $g(t) = \text{sinc}^2(5t)$. One sinc has bandwidth 2.5 Hz. Squaring doubles the support limit, so this signal has bandwidth $B = 5$ Hz. Its Nyquist rate is 10 samples/s.

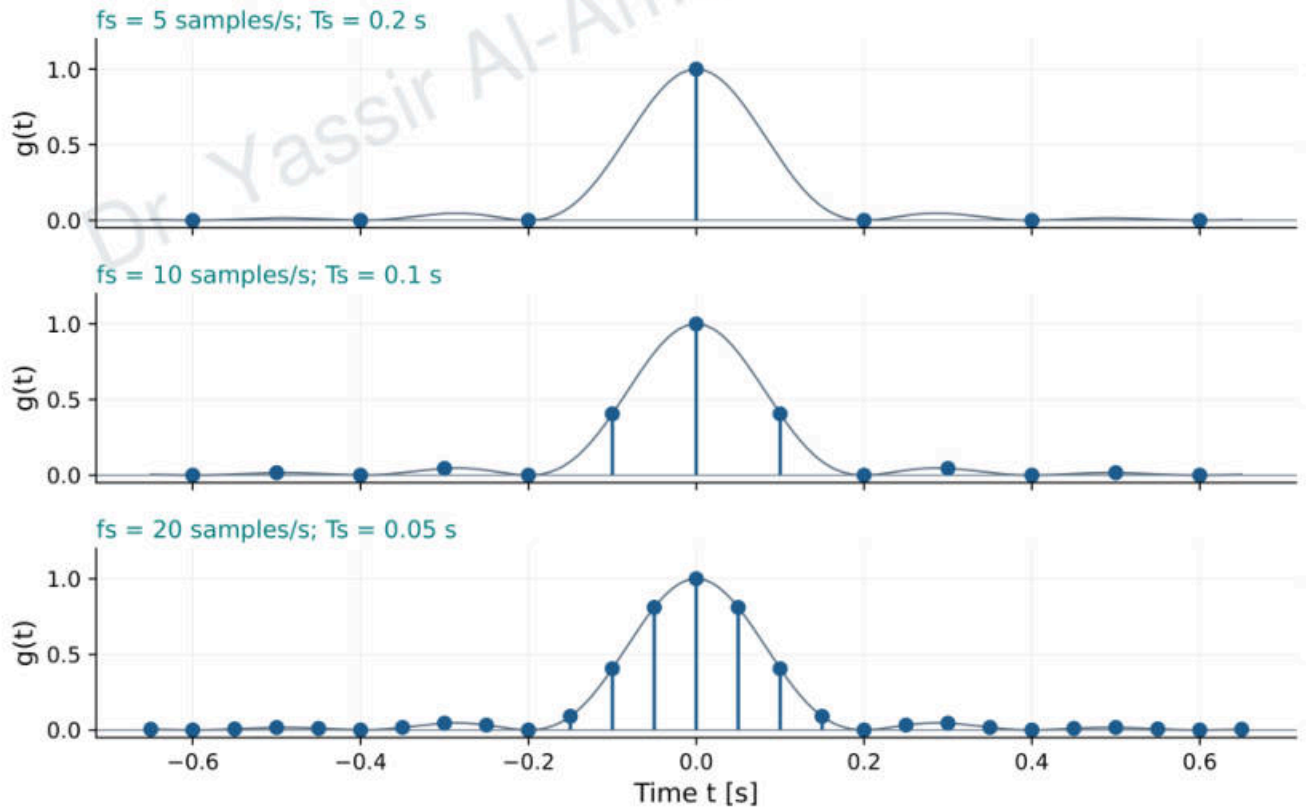


Figure 2.20: The dots show the measured values. The stems show ideal sample weights.

At 5 samples/s: the measurements miss the nonzero parts between the zeros. Only the centre sample is nonzero. Unique recovery is not possible for general messages with this bandwidth. **At 10 samples/s:** more nonzero values are measured. This particular signal can be recovered ideally because its spectrum is zero where the copies touch. **At 20 samples/s:** there are more measurements and a wider gap between spectral copies. The original message itself has not changed.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.31

See the same sampling rates in frequency

Overlap is easier to see in the spectrum.

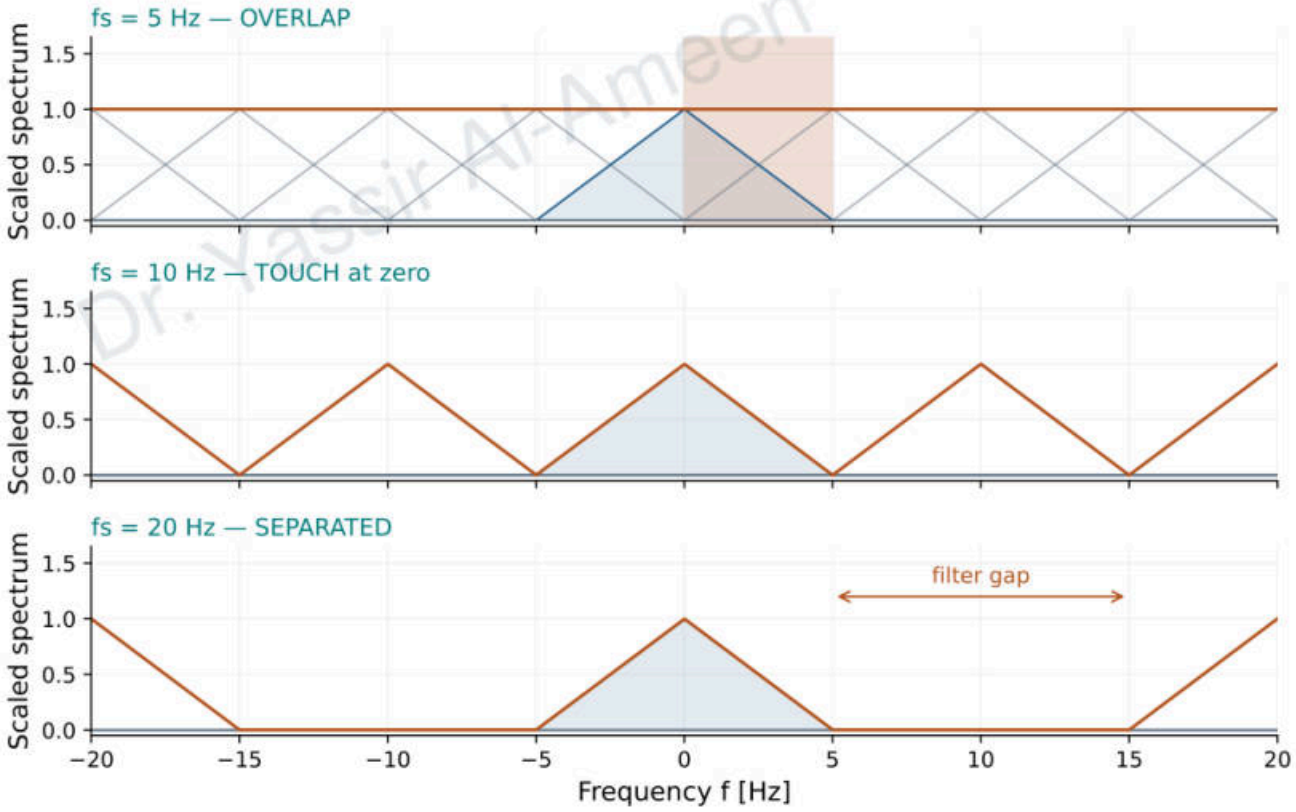


Figure 2.21: At 5 Hz the copies overlap. At 10 Hz they touch. At 20 Hz they are separated. Heights are scaled for comparison.

The message spectrum is a triangle from -5 to 5 Hz. The shaded triangle is the central copy. Orange shows the sum. All curves are divided by $f_s/5$ to compare their shapes. The other copies move further away as the sampling rate increases. **At 5 Hz:** the triangles overlap and add to a flat spectrum. A low-pass filter cannot recover the original triangle. **At 10 Hz:** the triangles touch at their zero-valued ends. An ideal filter can keep the central triangle. **At 20 Hz:** a gap separates the triangles. This makes practical filtering easier. For calculations, the triangular spectrum is

$$G(f) = \frac{1}{5}\,\mathrm{tri}(f/5), \quad \mathrm{tri}(u) = \max(1 - |u|, 0).$$

Ideal impulse sampling repeats it and scales it by f_s:

$$G_s(f) = f_s \sum_n G(f - nf_s).$$

ANALOG PULSE MODULATION / 2.32

What comes out of the filter?

Separate the recovered shape from its amplitude.

Use the same message $g(t) = \text{sinc}^2(5t)$. Pass the sampled signal through an ideal LPF with bandwidth 5 Hz and gain one.

Example 2.21 / Compare the three outputs

Sampling at 5 Hz. All samples except the centre are zero. The sampled signal is one impulse, so its spectrum is one. The filter keeps a rectangle from -5 to 5 Hz:

$$Y(f) = \text{rect}(f/10), \qquad y(t) = 10\,\text{sinc}(10t).$$

This is not the original sinc-squared message. Aliasing has changed its shape.

Sampling at 10 Hz. The filter keeps the central triangle, but it is ten times larger than the original spectrum. The output is $10g(t)$. A gain of $1/10$ restores $g(t)$.

Sampling at 20 Hz. The output is $20g(t)$. A gain of $1/20$ restores $g(t)$.

Which recovery gain?. For ideal impulse sampling, use $1/f_s$. For natural rectangular-gate sampling, use $1/d$. These are different sampling models.

Short practice

A message has $B = 2$ kHz. Natural sampling uses $f_s = 6$ kHz and $d = 1/4$. Find the recovery gain and a filter cutoff. **Answer:** gain $1/d = 4$. The filter gap is 2 to 4 kHz, so 3 kHz is a suitable ideal cutoff.

A PWM system uses $\tau_0 = 40\,\mu$s and $k_w = 20\,\mu$s/V. Find the width for a 2 V sample. **Answer:** extra width $= 20(2) = 40\,\mu$s. Total width $= 40 + 40 = 80\,\mu$s.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.33

Question bank / Basic ideas

Easy questions with clear, worked answers.

QUESTION 1

What changes in PAM, PWM and PPM?

Solution

PAM: pulse height changes.

PWM: pulse width changes.

PPM: pulse position changes.

Answer: PWM and PPM have fixed pulse height; PPM also has fixed width.

QUESTION 2

What is the difference between natural and flat-top PAM?

Solution

Natural PAM: the pulse top follows the message while the gate is ON.

Flat-top PAM: the pulse holds one sample value and has a horizontal top.

Answer: Natural PAM follows the input; flat-top PAM holds a value.

QUESTION 3

Explain sample, hold and reset.

Solution

Sample: measure the input voltage.

Hold: keep that measured voltage for a short time.

Reset: return the output to zero for return-to-zero PAM.

Answer: A zero-order hold instead keeps the sample until the next sample.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.34

Question bank / Sampling calculations

Easy questions with clear, worked answers.

QUESTION 4

A message has a highest frequency of 2 kHz. Find its Nyquist rate.

Solution

Double the highest message frequency.

$$f_N = 2B$$

Substitute the bandwidth.

$$f_N = 2(2) = 4\,\text{ksample/s}$$

Answer: The Nyquist rate is 4 ksample/s. Use a rate above this boundary in practice.

QUESTION 5

The sampling frequency is 10 kHz. Find the sampling period.

Solution

Use the inverse of the sampling frequency.

$$T_s = \frac{1}{f_s}$$

Convert kHz to Hz and substitute.

$$T_s = \frac{1}{10000} = 0.0001\,\text{s}$$

Answer: $T_s = 100\,\mu\text{s}$.

QUESTION 6

Find the Nyquist rate for $x(t) = 3\cos(200\pi t) + 2\sin(600\pi t)$.

Solution

Convert angular frequency to hertz.

$$f = \frac{\omega}{2\pi}$$

Find the two frequencies.

$$f_1 = 100\,\text{Hz}, \qquad f_2 = 300\,\text{Hz}$$

Double the larger frequency.

$$f_N = 2(300) = 600\,\text{samples/s}$$

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ANALOG PULSE MODULATION / 2.35

Question bank / Aliasing and filtering

Easy questions with clear, worked answers.

QUESTION 7

Why is an anti-alias filter needed before sampling?

Solution

Remove unwanted high-frequency components before taking samples.

Otherwise, they may overlap the wanted spectrum after sampling.

Answer: Filtering first helps prevent aliasing; filtering afterward cannot generally undo it.

QUESTION 8

A 700 Hz cosine is sampled at 1000 samples/s. Find its lower-frequency alias.

Solution

Subtract one sampling frequency.

$$f - f_s = 700 - 1000 = -300\,\text{Hz}$$

For a cosine, positive and negative frequencies give the same samples.

$$f_{\text{alias}} = |-300| = 300\,\text{Hz}$$

Answer: The samples also fit a 300 Hz cosine.

QUESTION 9

Use $B = 2$ kHz and $f_s = 8$ kHz. Choose a reconstruction-filter cutoff.

Solution

Find where the next spectral copy begins.

$$f_s - B = 8 - 2 = 6\,\text{kHz}$$

Choose a cutoff inside the gap.

$$2 < f_c < 6\,\text{kHz}$$

Answer: An ideal cutoff of 4 kHz passes the message and rejects the next copy.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.36

Question bank / Natural PAM

Easy questions with clear, worked answers.

QUESTION 10

A gate has $T_s = 100\,\mu$s and width $\tau = 25\,\mu$s. Find its duty factor and recovery gain.

Solution

Divide ON time by period.

$$d = \frac{\tau}{T_s} = \frac{25}{100} = 0.25$$

Invert the duty factor.

$$G = \frac{1}{d} = \frac{1}{0.25} = 4$$

Answer: Duty factor: 25%. Recovery gain: 4, when spectral copies are separated.

QUESTION 11

For centred rectangular gates with $d = 1/2$, find c_0 and the first missing spectral copy.

Solution

Use the gate coefficient.

$$c_n = d\,\mathrm{sinc}(nd)$$

At the centre, sinc(0) equals one.

$$c_0 = \tfrac{1}{2}$$

The first zero occurs when the sinc argument is one.

$$n/2 = 1 \quad \Rightarrow \quad n = 2$$

Answer: The first missing copies are at $\pm 2f_s$. Higher copies may still exist.

QUESTION 12

State the job of each block in a natural-PAM system.

Solution

Anti-alias filter: limits input bandwidth.

Sampling switch: passes the message during each ON interval.

Receiver LPF: selects the central spectrum.

Gain $1/d$: restores the message amplitude.

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.37

Question bank / Hold and reconstruction

Easy questions with clear, worked answers.

QUESTION 13

Let $h(t) = 1$ for $0 \le t < 10\,\mu$s and zero otherwise. Find $2\delta(t - 30\,\mu\text{s}) * h(t)$.

Solution

A shifted impulse shifts the hold pulse.

$$g(t) = 2h(t - 30\,\mu\text{s})$$

Find the pulse end.

$$30 + 10 = 40\,\mu\text{s}$$

Answer: A height-2 rectangle starts at $30\,\mu$s and ends at $40\,\mu$s.

QUESTION 14

What is aperture effect? For a hold width of $20\,\mu$s, find the first spectral null.

Solution

A finite flat-top hold reduces different message frequencies by different amounts. This amplitude droop is the aperture effect.

The first null of the rectangular hold occurs at:

$$f_{\text{null}} = \frac{1}{\tau}$$

Substitute the duration in seconds.

$$f_{\text{null}} = \frac{1}{20 \times 10^{-6}} = 50\,\text{kHz}$$

QUESTION 15

How does ideal sinc interpolation recover the sample value at a sampling instant?

Solution

At $t = kT_s$, each sinc argument is an integer.

$$\text{sinc}\left(\frac{kT_s - nT_s}{T_s}\right) = \text{sinc}(k - n)$$

When $n = k$: the factor is one.

When $n \neq k$: the factor is zero.

Answer: Only the matching sample remains. Full recovery requires the sampling and bandlimiting conditions.

ANALOG PULSE MODULATION / 2.38

Question bank / Pulse-width modulation

Easy questions with clear, worked answers.

QUESTION 16

Use $\tau_k = 50 + 20x$ microseconds, where x is in volts. Find the widths for $x = -1, 0, +1$ V.

Solution

For a negative sample:

$$\tau_k = 50 + 20(-1) = 30\,\mu\text{s}$$

For a zero sample:

$$\tau_k = 50 + 20(0) = 50\,\mu\text{s}$$

For a positive sample:

$$\tau_k = 50 + 20(1) = 70\,\mu\text{s}$$

QUESTION 17

A ramp rises from 0 to 4 V in $100\,\mu$s. The control voltage is 2 V. Find the PWM width.

Solution

Use the fraction of the ramp reached by the control voltage.

$$\tau_k = T_s \frac{v_k}{V_r}$$

Substitute the values.

$$\tau_k = 100 \frac{2}{4} = 50\,\mu\text{s}$$

Answer: The comparator stays high for $50\,\mu$s.

QUESTION 18

A centred PWM pulse has centre $60\,\mu$s and width $40\,\mu$s. Find its two edges.

Solution

Find half the width.

$$\tau_k/2 = 40/2 = 20\,\mu\text{s}$$

Subtract it to find the start.

$$t_{\text{start}} = 60 - 20 = 40\,\mu\text{s}$$

Add it to find the end.

$$t_{\text{end}} = 60 + 20 = 80\,\mu\text{s}$$

Dr. Yassir Al-Ameen

ANALOG PULSE MODULATION / 2.39

Question bank / Pulse-position modulation

Easy questions with clear, worked answers.

QUESTION 19

How is PPM generated from fixed-leading-edge PWM?

Solution

Detect the moving **falling edge** of each PWM pulse.

Trigger a short, fixed-width pulse at that edge.

Keep its height and width constant.

Answer: A wider PWM pulse produces a later PPM pulse.

QUESTION 20

A PPM pulse arrives $70\ \mu$s after its reference. Use $\tau_0 = 50\ \mu$s and $k_w = 20\ \mu$s/V. Find the sample.

Solution

Write the inverse mapping.

$$x = \frac{d_k - \tau_0}{k_w}$$

Remove the fixed delay.

$$d_k - \tau_0 = 70 - 50 = 20\ \mu\text{s}$$

Divide by the sensitivity.

$$x = \frac{20}{20} = 1\ \text{V}$$

QUESTION 21

How does a set-reset flip-flop convert PPM into PWM?

Solution

The slot reference **sets** the output high.

The received PPM leading edge **resets** it low.

The time between these events becomes the output pulse width.

Answer: The PWM width equals the PPM delay. An accurate slot reference is essential.

ANALOG PULSE MODULATION / 2.40

Question bank / Noise and bandwidth

Easy questions with clear, worked answers.

QUESTION 22

A small disturbance is 0.02 V and the rising-edge slope is 0.1 V/μs. Find the approximate timing error.

Solution

Use the local linear approximation.

$$\Delta t \approx -\frac{v_n}{q}$$

Substitute the disturbance and slope.

$$\Delta t \approx -\frac{0.02}{0.1} = -0.2\,\mu\text{s}$$

Answer: The crossing occurs about 0.2 μs earlier. Limiting pulse height does not remove all timing error.

QUESTION 23

Use normalized sinc. Find the Nyquist rates for $\text{sinc}(100t)$ **and** $\text{sinc}^2(100t)$**.**

Solution

For one sinc, the bandwidth is half its time-scaling factor.

$$B = 100/2 = 50\,\text{Hz}, \qquad f_N = 2B = 100\,\text{samples/s}$$

Squaring multiplies two signals, so the bandwidths add.

$$B = 50 + 50 = 100\,\text{Hz}$$

Double the new bandwidth.

$$f_N = 200\,\text{samples/s}$$

QUESTION 24

Two signals fill bandwidths of 2 and 3 kHz. Find the Nyquist rate of their product.

Solution

Multiplication in time adds the spectral support limits.

$$B_y = B_1 + B_2$$

Add the bandwidths.

$$B_y = 2 + 3 = 5\,\text{kHz}$$

Double the output bandwidth.

$$f_{N,y} = 2(5) = 10\,\text{ksample/s}$$

ANALOG PULSE MODULATION / 2.41

Question bank / Read and draw the pulses

Easy questions with clear, worked answers.

QUESTION 25

Draw PAM, PWM and PPM for samples $-1, 0, +1$ V at $0, 100, 200\ \mu$s. Use PAM width $20\ \mu$s, PWM width $50 + 20x$ microseconds, and PPM width $10\ \mu$s.

Solution

Use the sample values as PAM heights.

$$-1, \quad 0, \quad +1\,\text{V}$$

Calculate the PWM widths in the three slots.

$$30, \quad 50, \quad 70\ \mu\text{s}$$

Add each width to its slot start to locate the PPM leading edge.

$$0 + 30 = 30, \quad 100 + 50 = 150, \quad 200 + 70 = 270\ \mu\text{s}$$

Add the fixed PPM width to locate its trailing edge.

$$40, \quad 160, \quad 280\ \mu\text{s}$$

Answer: The zero-height PAM pulse is a valid zero sample, not a missing sample.

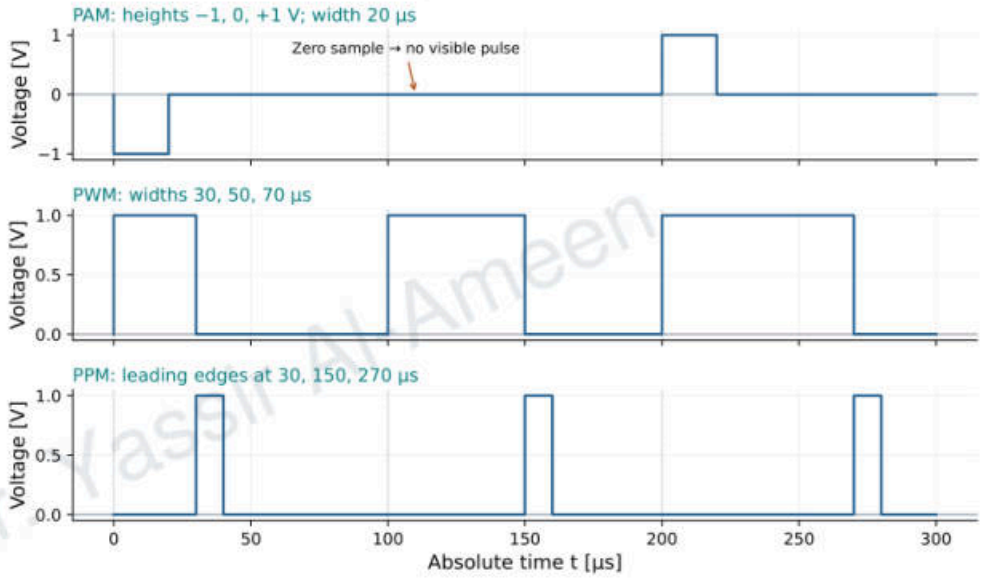


Figure 2.22: Answer to Q25. Dotted lines show slot starts; each PPM pulse starts at the corresponding PWM falling edge.